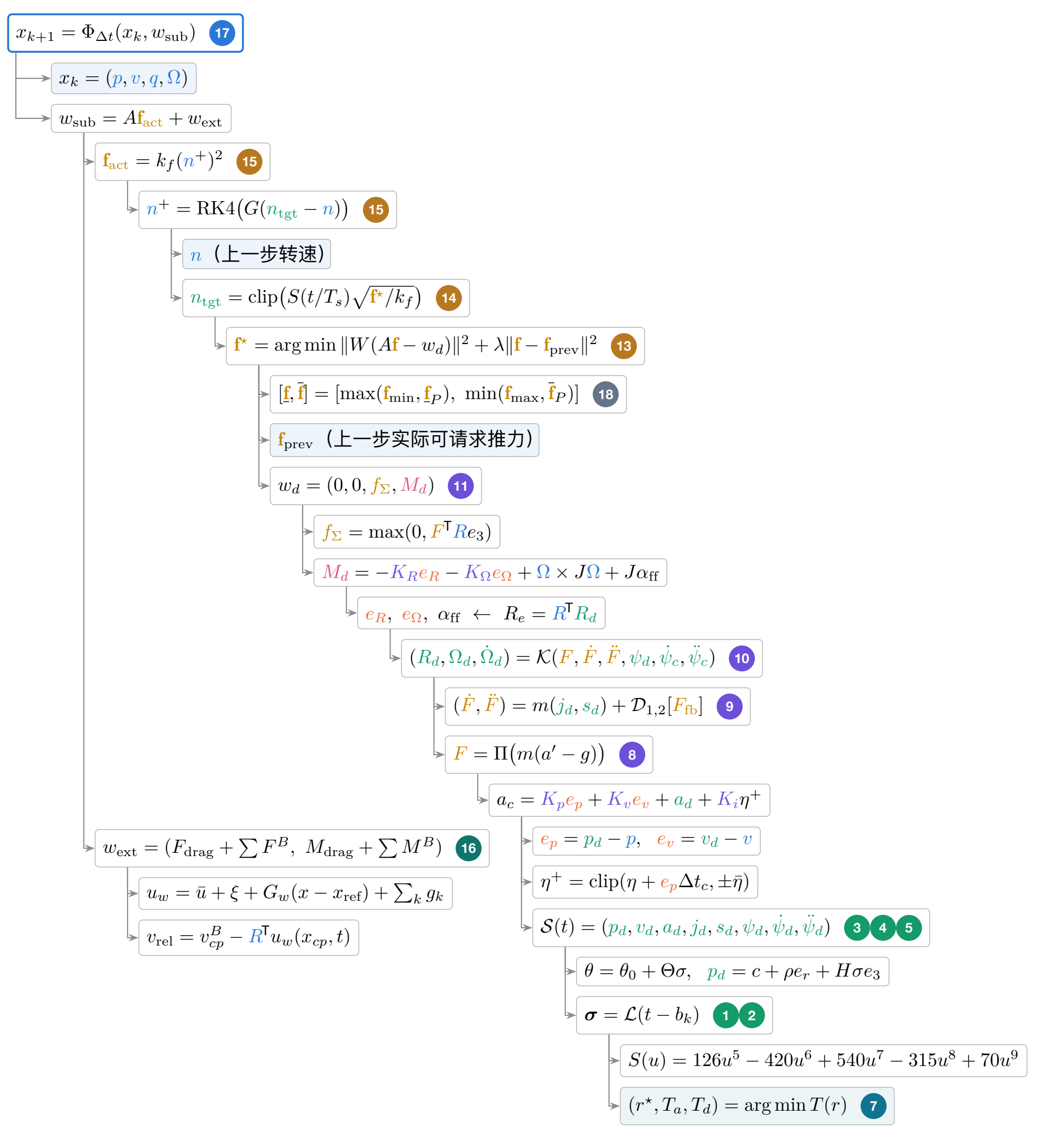
isaac-drone-sim 纯数学计算过程·公式逻辑树

全部为符号公式，不代入任何配置数值；每条公式都与 `isaac_drone/` 源码逐条核对。
模块内细箭头 = 「由上游公式代入得到」；模块间粗箭头 = 跨模块传递的量；虚线 = 反馈到下一步 / 退化分支。

总逻辑树：从 x_{k+1} 倒推它依赖的公式

父 → 子 = “由谁算出”；浅色叶子 = 上一步状态 / 复位时规划



- $x_{k+1} = \Phi_{\Delta t}(x_k, w_{\text{sub}})$ ⑰
  - $x_k = (p, v, q, \Omega)$
  - $w_{\text{sub}} = A\mathbf{f}_{\text{act}} + w_{\text{ext}}$
    - $\mathbf{f}_{\text{act}} = k_f (n^+)^2$ ⑮
      - $n^+ = \text{RK4}(G(n_{\text{tgt}} - n))$ ⑮
        - n（上一步转速）
        - $n_{\text{tgt}} = \text{clip}(S(t/T_s)\sqrt{\mathbf{f}^\star/k_f})$ ⑭
          - $\mathbf{f}^\star = \arg\min \|W(A\mathbf{f} - w_d)\|^2 + \lambda\|\mathbf{f} - \mathbf{f}_{\text{prev}}\|^2$ ⑬
            - $[\underline{\mathbf{f}}, \bar{\mathbf{f}}] = [\max(\mathbf{f}_{\min}, \underline{\mathbf{f}}_P),\ \min(\mathbf{f}_{\max}, \bar{\mathbf{f}}_P)]$ ⑱
            - $\mathbf{f}_{\text{prev}}$（上一步实际可请求推力）
            - $w_d = (0, 0, f_\Sigma, M_d)$ ⑪
              - $f_\Sigma = \max(0, F^\top R e_3)$
              - $M_d = -K_R e_R - K_\Omega e_\Omega + \Omega \times J\Omega + J\alpha_{\text{ff}}$
                - $e_R,\ e_\Omega,\ \alpha_{\text{ff}} \leftarrow R_e = R^\top R_d$
                  - $(R_d, \Omega_d, \dot\Omega_d) = \mathcal{K}(F, \dot F, \ddot F, \psi_d, \dot\psi_c, \ddot\psi_c)$ ⑩
                    - $(\dot F, \ddot F) = m(j_d, s_d) + \mathcal{D}_{1,2}[F_{\text{fb}}]$ ⑨
                    - $F = \Pi(m(a' - g))$ ⑧
                      - $a_c = K_p e_p + K_v e_v + a_d + K_i \eta^+$
                        - $e_p = p_d - p,\quad e_v = v_d - v$
                        - $\eta^+ = \text{clip}(\eta + e_p \Delta t_c, \pm\bar\eta)$
                        - $\mathcal{S}(t) = (p_d, v_d, a_d, j_d, s_d, \psi_d, \dot\psi_d, \ddot\psi_d)$ ③④⑤
                          - $\theta = \theta_0 + \Theta\sigma,\quad p_d = c + \rho e_r + H\sigma e_3$
                          - $\boldsymbol\sigma = \mathcal{L}(t - b_k)$ ①②
                            - $S(u) = 126u^5 - 420u^6 + 540u^7 - 315u^8 + 70u^9$
                            - $(r^\star, T_a, T_d) = \arg\min T(r)$ ⑦
    - $w_{\text{ext}} = (F_{\text{drag}} + \sum F^B,\ M_{\text{drag}} + \sum M^B)$ ⑯
      - $u_w = \bar u + \xi + G_w(x - x_{\text{ref}}) + \sum_k g_k$
      - $v_{\text{rel}} = v^B_{cp} - R^\top u_w(x_{cp}, t)$

总公式：一个物理步 $k \to k+1$

$$x_{k+1} = \Phi_{\Delta t}\Big(x_k,\ A\,k_f \odot \Big[\mathcal{M}_{\Delta t}\Big(n_k,\ \mathcal{R}\Big(\mathcal{A}\big(\mathcal{C}(x_k,\ \mathcal{G}(\mathcal{L}(t_k))),\ [\underline{\mathbf{f}}, \bar{\mathbf{f}}]\big),\ t_k\Big)\Big)\Big]^2 + w_{\text{ext}}(x_k, t_k)\Big)$$

逐层展开（右侧圆章 = 模块编号）

$x_{k+1} = \Phi_{\Delta t}(x_k,\ w_{\text{sub}}),\quad w_{\text{sub}} = A\mathbf{f}_{\text{act}} + w_{\text{ext}}(x_k, t_k)$ ⑰ ⑯

$\mathbf{f}_{\text{act}} = k_f \odot (n^+)^2,\quad n^+ = \mathcal{M}_{\Delta t}(n_k, n_{\text{tgt}})$ ⑮

$n_{\text{tgt}} = \mathcal{R}(\mathbf{f}^\star, t_k) = \text{clip}\Big(S\big(\tfrac{t_k}{T_s}\big)\sqrt{\mathbf{f}^\star/k_f},\ n_{\min}, n_{\max}\Big)$ ⑭

$\mathbf{f}^\star = \mathcal{A}(w_d;\ [\underline{\mathbf{f}}, \bar{\mathbf{f}}],\ \mathbf{f}_{\text{prev}})$ ⑬ ⑱

$w_d = \mathcal{C}(x_k,\ \mathcal{S}(t_k))$（仅 $k \bmod N_c = 0$ 时重算） ⑧—⑪ 或 ⑫

$\mathcal{S}(t) = \mathcal{G}(\boldsymbol\sigma(t)),\quad \boldsymbol\sigma = \mathcal{L}(t;\ T$ 或 $(r^\star, T_a, T_d))$ ③—⑤ ① ② ⑦

记号与图例

符号颜色　实际 p, v, R, Ω, n　期望/参考 $p_d, R_d, \Omega_d, n_{\text{tgt}}$　误差 e_p, e_R, ϱ
推力 $F, f_\Sigma, \mathbf{f}$　力矩 M_d　增益 K_p, K_R

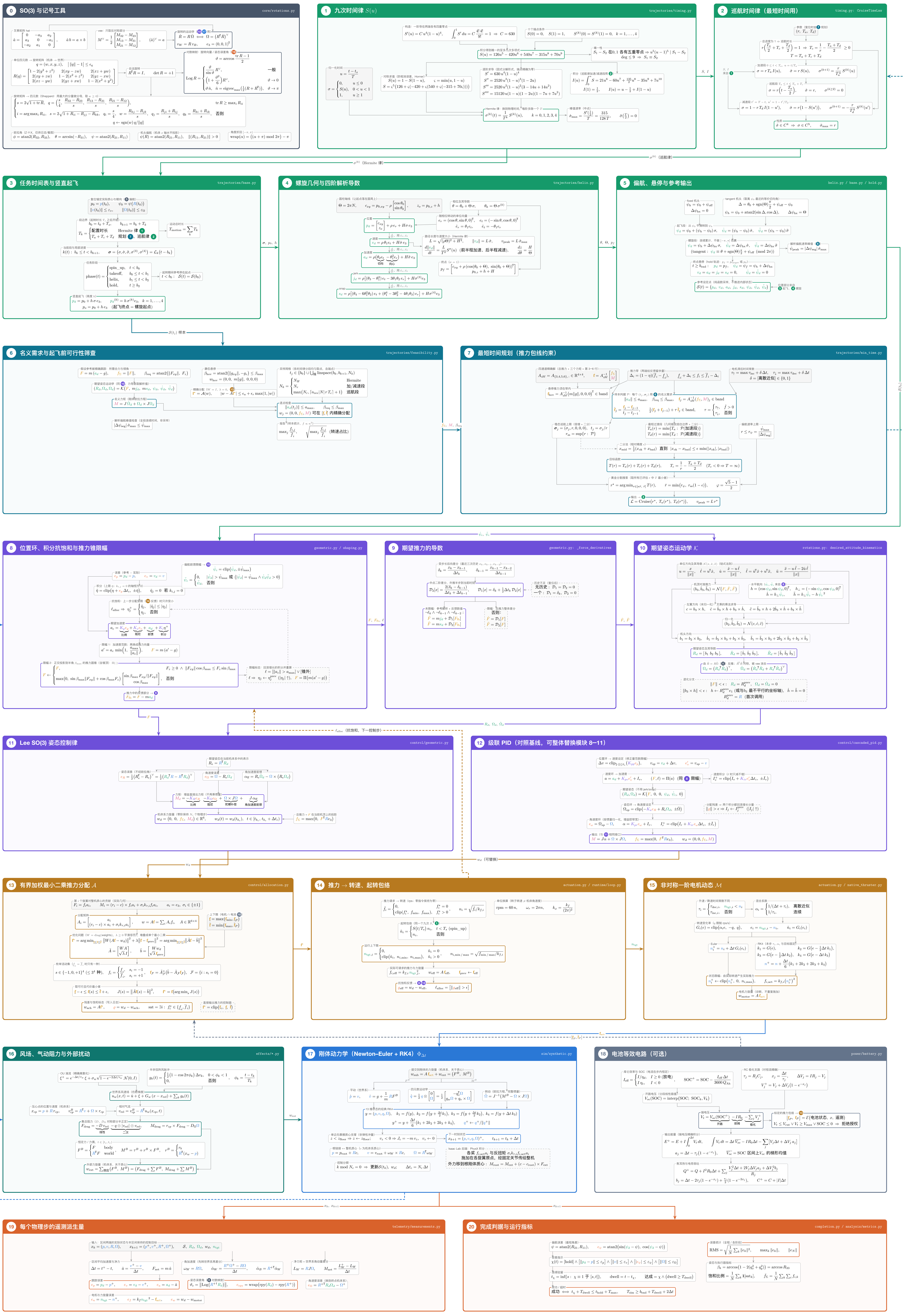
坐标　世界系 W（z 向上）、机体系 B（$+z$ 为推力方向）；$R: B \to W$；$e_3 = (0,0,1)^\top$；g 为世界系重力
时间　t 绝对仿真时间；Δt 物理步长；$\Delta t_c = N_c\Delta t$ 控制周期；上标 $+$ 为下一步；$x^{(k)}$ 为 k 阶导数
区分　ρ 螺旋半径；r 巡航进度速率；r_i 旋翼位置；c 机体系质心；f_Σ 总推力；$\mathbf{f} = (f_1, \dots, f_4)^\top$
运算　$\text{clip}(x, a, b) = \min(\max(x, a), b)$；$[\cdot]$ 艾佛森括号（真为 1）；$\odot$ 逐元素乘；A^+ 伪逆

→ 公式代入（模块内）　⇢ 退化 / 回滚分支
➞ 跨模块数据流（同一步）　⇢ 反馈到下一步 / 复位时一次性
普通公式　模块输出　输入 / 定义　条件 / 退化分支

一个物理步的执行顺序（runtime/loop.py）

复位时一次　$x_0 \to$ ③ $(p_0, \psi_0) \to$ ⑦ $(r^\star, T_a, T_d) \to$ ② $\to$ ⑥ 筛查

每个物理步　$x_k \xrightarrow{k \bmod N_c = 0} \mathcal{S}(t_k)$ ③—⑤ $\to w_d$ ⑧—⑪ $\to \mathbf{f}^\star$ ⑬ $\to n_{\text{tgt}}$ ⑭ $\to \ell_{\text{alloc}} \to$ ⑧
$\to w_{\text{ext}}$ ⑯ $\to n^+, \mathbf{f}_{\text{act}}$ ⑮ $\to x_{k+1}$ ⑰ $\to$ 遥测 ⑲ $\to$ 完成判据 ⑳ $\to$ 电池 ⑱



0 SO(3) 与记号工具

`core/rotations.py`

叉乘矩阵 hat: $\hat a = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix},\quad \hat a\, b = a \times b$

vee：只取反对称部分: $M^\vee = \frac{1}{2}\begin{bmatrix} M_{32} - M_{23} \\ M_{13} - M_{31} \\ M_{21} - M_{12} \end{bmatrix},\quad (\hat a)^\vee = a$

旋转的运动学（⑩ ⑰ 用）: $\dot R = R\hat\Omega \iff \Omega = (R^\top \dot R)^\vee$，$v_W = R\, v_B,\quad e_3 = (0,0,1)^\top$

单位四元数 → 旋转矩阵（机体 → 世界）: $q = (w, x, y, z),\quad \big|\|q\| - 1\big| \le \epsilon_q$

$$R(q) = \begin{bmatrix} 1 - 2(y^2 + z^2) & 2(xy - zw) & 2(xz + yw) \\ 2(xy + zw) & 1 - 2(x^2 + z^2) & 2(yz - xw) \\ 2(xz - yw) & 2(yz + xw) & 1 - 2(x^2 + y^2) \end{bmatrix}$$

合法旋转: $R^\top R = I,\quad \det R = +1$

对数映射：旋转向量 / 姿态误差角（⑲）: $\vartheta = \arccos\frac{\operatorname{tr} R - 1}{2}$

$$\operatorname{Log} R = \begin{cases} \frac{\vartheta}{\sin\vartheta} R^\vee, & \text{一般} \\ \big(1 + \frac{\vartheta^2}{6}\big) R^\vee, & \vartheta \to 0 \\ \vartheta\,\hat n,\quad \hat n = \text{eigvec}_{\max}\big(\tfrac{1}{2}(R + R^\top)\big), & \vartheta \to \pi \end{cases}$$

旋转矩阵 → 四元数（Shepperd：用最大的分量做分母，取 $w \ge 0$）:

$$\begin{cases} s = 2\sqrt{1 + \operatorname{tr} R},\quad q = \Big(\frac{s}{4}, \frac{R_{32} - R_{23}}{s}, \frac{R_{13} - R_{31}}{s}, \frac{R_{21} - R_{12}}{s}\Big), & \operatorname{tr} R \ge \max_i R_{ii} \\ i = \arg\max_i R_{ii},\quad s = 2\sqrt{1 + R_{ii} - R_{jj} - R_{kk}},\quad q_i = \frac{s}{4},\quad w = \frac{R_{kj} - R_{jk}}{s},\quad q_j = \frac{R_{ji} + R_{ij}}{s},\quad q_k = \frac{R_{ki} + R_{ik}}{s}, & \text{否则} \end{cases}$$

$q \leftarrow \text{sgn}(w)\, q/\|q\|$

欧拉角（Z-Y-X，仅供日志/看图）: $\phi = \text{atan2}(R_{32}, R_{33}),\quad \theta = \arcsin(-R_{31}),\quad \psi = \text{atan2}(R_{21}, R_{11})$

机头偏航（机体 x 轴水平投影）: $\psi(R) = \text{atan2}(R_{21}, R_{11}),\quad \|(R_{11}, R_{21})\| > 0$

角度折回 $[-\pi, \pi)$: $\text{wrap}(\alpha) = \big((\alpha + \pi) \bmod 2\pi\big) - \pi$

1 九次时间律 $S(u)$

`trajectories/timing.py`

构造：一阶导在两端各有四重零点: $S'(u) = C\,u^4(1-u)^4,\quad \int_0^1 S'\,du = C\,\frac{4!\,4!}{9!} = 1 \Rightarrow C = 630$

十个端点条件: $S(0) = 0,\quad S(1) = 1,\quad S^{(k)}(0) = S^{(k)}(1) = 0,\quad k = 1, \dots, 4$

积分得到唯一的至多九次多项式: $S(u) = 126u^5 - 420u^6 + 540u^7 - 315u^8 + 70u^9$

唯一性: $S_1 - S_2$ 在 0, 1 各有五重零点 $\Rightarrow u^5(u-1)^5 \mid S_1 - S_2$，$\deg \le 9 \Rightarrow S_1 \equiv S_2$

归一化时间: $u = \frac{t - t_s}{T}$，$\sigma = \begin{cases} 0, & u \le 0 \\ S(u), & 0 < u < 1 \\ 1, & u \ge 1 \end{cases}$

对称求值（防相消误差，Horner）: $S(u) = 1 - S(1-u),\quad \varsigma = \min(u, 1-u)$，$S = \varsigma^5\big(126 + \varsigma(-420 + \varsigma(540 + \varsigma(-315 + 70\varsigma)))\big)$

逐阶求导（因式分解形式，端点精确为零）:
$S' = 630\,u^4(1-u)^4$
$S'' = 2520\,u^3(1-u)^3(1-2u)$
$S''' = 2520\,u^2(1-u)^2(3 - 14u + 14u^2)$
$S'''' = 15120\,u(1-u)(1-2u)(1 - 7u + 7u^2)$

积分（巡航律加速/减速段用 ②）: $I(u) = \int_0^u S = 21u^6 - 60u^7 + \frac{135}{2}u^8 - 35u^9 + 7u^{10}$，$I(1) = \frac{1}{2},\quad I(u) = u - \frac{1}{2} + I(1-u)$

Hermite 律：换到物理时间，每阶多除一个 T: $\sigma^{(k)}(t) = \frac{1}{T^k} S^{(k)}(u),\quad k = 0, 1, 2, 3, 4$

峰值速率（中点）: $\dot\sigma_{\max} = \frac{S'(\frac{1}{2})}{T} = \frac{315}{128\,T},\quad \ddot\sigma\big(\tfrac{T}{2}\big) = 0$

$S,\ I$ → ②

$\sigma^{(k)}$（Hermite 律） → ③ ④

2 巡航时间律（最短时间用）

`timing.py: CruiseTimeLaw`

参数（复位时由 ⑦ 规划）: $(r,\ T_a,\ T_d)$

总进度为 1 ⇒ 巡航时长: $r\Big(\frac{T_a}{2} + T_c + \frac{T_d}{2}\Big) = 1 \Rightarrow T_c = \frac{1}{r} - \frac{T_a + T_d}{2} \ge 0$，$T = T_a + T_c + T_d$

$S,\ I$ 来自 ①

加速段 $0 \le t < T_a,\ u = t/T_a$: $\sigma = r\,T_a\,I(u),\quad \dot\sigma = r\,S(u),\quad \sigma^{(k+1)} = \frac{r}{T_a^k} S^{(k)}(u)$

巡航段 $T_a \le t \le T_a + T_c$: $\sigma = r\Big(t - \frac{T_a}{2}\Big),\quad \dot\sigma = r,\quad \sigma^{(k \ge 2)} = 0$

减速段 $t' = T - t,\ u' = 1 - t'/T_d$: $\sigma = 1 - r\,T_d\,I(1 - u'),\quad \dot\sigma = r\big(1 - S(u')\big),\quad \sigma^{(k+1)} = -\frac{r}{T_d^k} S^{(k)}(u')$

性质: $\dot\sigma \in C^4 \Rightarrow \sigma \in C^5,\quad \dot\sigma_{\max} = r$

$\sigma^{(k)}$（巡航律） → ③ ④

3 任务时间表与竖直起飞

`trajectories/base.py`

复位锚定实际质心与朝向（⓪ 偏航）: $p_0 = p(t_0),\quad \psi_0 = \psi(R(t_0))$，$\|v(t_0)\| \le \varepsilon_v,\quad \|\Omega(t_0)\| \le \varepsilon_\Omega$

段边界（起转时长 T_s 之后开始）: $b_0 = t_0 + T_s,\quad b_{k+1} = b_k + T_k$，$T_k = \begin{cases} \text{配置时长} & \text{Hermite 律 ①} \\ T_a + T_c + T_d & \text{规划 ⑦，巡航律 ②} \end{cases}$

运动总时长: $T_{\text{motion}} = \sum_k T_k$

当前段与局部进度: $k(t):\ b_k \le t < b_{k+1},\quad \boldsymbol\sigma = (\sigma, \dot\sigma, \ddot\sigma, \sigma^{(3)}, \sigma^{(4)}) = \mathcal{L}_k(t - b_k)$

任务阶段: $\text{phase}(t) = \begin{cases} \text{spin\_up}, & t < b_0 \\ \text{takeoff}, & b_0 \le t < b_1 \\ \text{helix}, & b_1 \le t < b_2 \\ \text{hold}, & t \ge b_2 \end{cases}$

起转期间参考停在起点: $t < b_0:\ \mathcal{S}(t) = \mathcal{S}(b_0)$

竖直起飞（高度 h）: $p_d = p_0 + h\,\sigma\,e_3,\quad p_d^{(k)} = h\,\sigma^{(k)} e_3,\quad k = 1, \dots, 4$，$p_o = p_0 + h\,e_3$（起飞终点 = 螺旋起点）

$\boldsymbol\sigma,\ p_0,\ h$ → ④

$\mathcal{S}(t_j)$ 样本 → ⑥

4 螺旋几何与四阶解析导数

`trajectories/helix.py`

圆柱轴线（让起点落在圆周上）: $\Theta = 2\pi N,\quad c_{xy} = p_{0,xy} - \rho\begin{bmatrix}\cos\theta_0 \\ \sin\theta_0\end{bmatrix},\quad z_o = p_{0,z} + h$

相位及其导数: $\theta = \theta_0 + \Theta\,\sigma,\quad \theta_k = \Theta\,\sigma^{(k)}$

随相位转动的单位向量: $e_r = (\cos\theta, \sin\theta, 0)^\top,\quad e_t = (-\sin\theta, \cos\theta, 0)^\top$，$\dot e_r = \theta_1 e_t,\quad \dot e_t = -\theta_1 e_r$

位置: $p_d = \begin{bmatrix} c_{xy} \\ z_o \end{bmatrix} + \rho\,e_r + H\sigma\,e_3$

$\frac{d}{dt}$，用 $\dot e_r, \dot e_t$

速度: $v_d = \rho\,\theta_1 e_t + H\dot\sigma\,e_3$

路径长度与速度大小（Hermite 律）: $L = \sqrt{(\rho\Theta)^2 + H^2},\quad \|v_d\| = L\,\dot\sigma,\quad v_{\text{peak}} = L\,\dot\sigma_{\max}$，$\frac{d\|v_d\|}{dt} = \frac{L}{T^2} S''(u)$（前半程加速、后半程减速），$\frac{dz}{d\theta} = \frac{H}{\Theta}$

加速度: $a_d = \rho\big(\underbrace{\theta_2 e_t}_{\text{切向}} \underbrace{- \theta_1^2 e_r}_{\text{向心}}\big) + H\ddot\sigma\,e_3$

jerk: $j_d = \rho\big[(\theta_3 - \theta_1^3)\,e_t - 3\theta_1\theta_2\,e_r\big] + H\sigma^{(3)} e_3$

snap: $s_d = \rho\big[(\theta_4 - 6\theta_1^2\theta_2)\,e_t + (\theta_1^4 - 3\theta_2^2 - 4\theta_1\theta_3)\,e_r\big] + H\sigma^{(4)} e_3$

终点（$\sigma = 1$）: $p_f = \begin{bmatrix} c_{xy} + \rho\,(\cos(\theta_0 + \Theta),\ \sin(\theta_0 + \Theta))^\top \\ p_{0,z} + h + H \end{bmatrix}$

$\theta,\ \Theta,\ p_f$ → ⑤

5 偏航、悬停与参考输出

`helix.py / base.py / hold.py`

fixed 机头: $\psi_h = \psi_0 + \psi_{\text{off}}$，$\Delta\psi_{\text{hx}} = 0$

tangent 机头（取离 ψ_0 最近的等价切向角）: $\Delta = \theta_0 + \text{sgn}(\Theta)\frac{\pi}{2} + \psi_{\text{off}} - \psi_0$，$\psi_h = \psi_0 + \text{atan2}(\sin\Delta, \cos\Delta),\quad \Delta\psi_{\text{hx}} = \Theta$

起飞段：从 ψ_0 平滑转到 ψ_h: $\psi_d = \psi_0 + (\psi_h - \psi_0)\,\sigma,\quad \dot\psi_d = (\psi_h - \psi_0)\,\dot\sigma,\quad \ddot\psi_d = (\psi_h - \psi_0)\,\ddot\sigma$

螺旋段：连续累计、不做 $[-\pi, \pi]$ 包裹: $\psi_d = \psi_h + \Delta\psi_{\text{hx}}\,\sigma,\quad \dot\psi_d = \Delta\psi_{\text{hx}}\,\dot\sigma,\quad \ddot\psi_d = \Delta\psi_{\text{hx}}\,\ddot\sigma$，（tangent：$\psi_d \equiv \theta + \text{sgn}(\Theta)\frac{\pi}{2} + \psi_{\text{off}} \pmod{2\pi}$）

解析偏航速率峰值（⑥）: $\dot\psi_{\text{peak}} = |\Delta\psi_{\text{seg}}|\,\dot\sigma_{\max}$

终点悬停（hold 轨迹：$p_f = p_{\text{target}}$ 或 p_0）: $t \ge b_{\text{end}}:\quad p_d = p_f,\quad \psi_d = \psi_f = \psi_h + \Delta\psi_{\text{hx}}$，$v_d = a_d = j_d = s_d = 0,\quad \dot\psi_d = \ddot\psi_d = 0$

参考设定点（纯函数采样，不推进内部状态）: $\mathcal{S}(t) = (p_d,\ v_d,\ a_d,\ j_d,\ s_d,\ \psi_d,\ \dot\psi_d,\ \ddot\psi_d)$

位置部分来自 ③ 起飞、④ 螺旋

$\mathcal{S}(t_k)$ → ⑧

6 名义需求与起飞前可行性筛查

`trajectories/feasibility.py`

假设参考被精确跟踪：所需合力与倾角: $F = m\,(a_d - g),\quad f_\Sigma = \|F\|,\quad \beta_{\text{req}} = \text{atan2}(\|F_{xy}\|,\ F_z)$

期望姿态运动学（同 ⑩，力导数取解析值）: $(R_d, \Omega_d, \dot\Omega_d) = \mathcal{K}(F,\ m j_d,\ m s_d,\ \psi_d,\ \dot\psi_d,\ \ddot\psi_d)$

名义力矩（刚体欧拉方程）: $M = J\,\dot\Omega_d + \Omega_d \times J\Omega_d$

解析偏航峰值检查（全段连续时间，非采样）: $|\Delta\psi_{\text{seg}}|\,\dot\sigma_{\max} \le \dot\psi_{\max}$

静态悬停: $\beta_{\text{hov}} = \text{atan2}(\|g_{xy}\|, -g_z) \le \beta_{\max}$，$w_{\text{hov}} = (0, 0,\ m\|g\|,\ 0, 0, 0)$

精确分配（$W = I,\ \lambda = 0$，⑬）: $\mathbf{f}^\star = \mathcal{A}(w),\quad |w - A\mathbf{f}^\star| \le \epsilon_a + \epsilon_r \max(1, |w|)$

采样网格（各时间律分段均匀取点，含端点）: $t_j \in \{b_0\} \cup \bigcup_{\text{段}} \text{linspace}(b_k, b_{k+1}, N_k)$，$N_k = \begin{cases} N_H & \text{Hermite} \\ N_r & \text{加/减速段} \\ \max(N_r, \lceil n_{\text{rev}} |N|\, r\, T_c \rceil + 1) & \text{巡航段} \end{cases}$

逐点检查: $\|a_d(t_j)\| \le a_{\max},\quad \beta_{\text{req}} \le \beta_{\max}$，$w_j = (0, 0, f_\Sigma, M)$ 可在 $[\underline{\mathbf{f}}, \bar{\mathbf{f}}]$ 内精确分配

报告（样本统计，$f \propto n^2$）: $\max_j \frac{f_{i,j}}{\bar f_i},\quad \sqrt{\max_j \frac{f_{i,j}}{\bar f_i}}$（转速占比）

$f_\Sigma,\ M,\ \beta_{\text{req}}$ → ⑦

7 最短时间规划（推力包线约束）

`trajectories/min_time.py`

四通道精确解（总推力 + 三个力矩 = 第 3–6 行）: $A_{zM} = A_{\{3,4,5,6\},:} \in \mathbb{R}^{4\times4},\quad \mathbf{f} = A_{zM}^{-1}\begin{bmatrix} f_\Sigma \\ M \end{bmatrix}$

推力带（两端给反馈留余量）: $\Delta_i = (1 - \eta)(\bar f_i - \underline{f}_i),\quad \underline{f}_i + \Delta_i \le f_i \le \bar f_i - \Delta_i$

电机滞后时间常数: $\tau_\uparrow = \max\tau_{\text{inc}} + \delta\,\Delta t,\quad \tau_\downarrow = \max\tau_{\text{dec}} + \delta\,\Delta t$，$\delta = [\text{离散近似}] \in \{0, 1\}$

悬停推力须在带内: $\mathbf{f}_{\text{hov}} = A_{zM}^{-1}(m\|g\|, 0, 0, 0)^\top \in \text{band}$

样本判据 $\mathcal{P}$：每个 $(t_j, \boldsymbol\sigma_j)$ 用 ⑥ 的名义需求: $\|a_d\| \le a_{\max},\quad \beta_{\text{req}} \le \beta_{\max},\quad \mathbf{f}_j = A_{zM}^{-1}(f_\Sigma, M)_j \in \text{band}$，$\dot{\mathbf{f}}_j = \frac{\mathbf{f}_j - \mathbf{f}_{j-1}}{t_j - t_{j-1}},\quad \frac{1}{2}(\mathbf{f}_j + \mathbf{f}_{j-1}) + \tau\,\dot{\mathbf{f}}_j \in \text{band},\quad \tau = \begin{cases} \tau_\uparrow, & \dot f > 0 \\ \tau_\downarrow, & \text{否则} \end{cases}$

稳态巡航上限（倍增 + 二分）: $\boldsymbol\sigma_j = (\sigma_j, r, 0, 0, 0),\quad t_j = \sigma_j / r$，$r_{\text{ss}} = \sup\{r:\ \mathcal{P}\}$

最短过渡（几何倍增括住边界 + 二分）: $T_a(r) = \min\{T_a:\ \mathcal{P}(\text{加速段})\}$，$T_d(r) = \min\{T_d:\ \mathcal{P}(\text{减速段})\}$

偏航速率上限: $r \le r_\psi = \frac{\dot\psi_{\max}}{|\Delta\psi_{\text{seg}}|}$

二分法（相对精度 ϵ）: $x_{\text{mid}} = \frac{1}{2}(x_{\text{ok}} + x_{\text{bad}})$ 直到 $|x_{\text{ok}} - x_{\text{bad}}| \le \epsilon \min(|x_{\text{ok}}|, |x_{\text{bad}}|)$

目标函数: $T(r) = T_a(r) + T_c(r) + T_d(r),\quad T_c = \frac{1}{r} - \frac{T_a + T_d}{2}\quad (T_c < 0 \Rightarrow T = \infty)$

黄金分割搜索（取所有已评估 r 中 T 最小者）: $r^\star = \arg\min_{r \in [\kappa\bar r,\ \bar r]} T(r),\quad \bar r = \min(r_\psi,\ r_{\text{ss}}(1 - \epsilon)),\quad \varphi = \frac{\sqrt5 - 1}{2}$

输出 → ②: $\mathcal{L} = \text{Cruise}(r^\star,\ T_a(r^\star),\ T_d(r^\star)),\quad v_{\text{peak}} = L\,r^\star$

$\dot\psi_c,\ \ddot\psi_c$ → ⑩

8 位置环、积分抗饱和与推力锥限幅

`geometric.py / shaping.py`

误差（参考 − 实际）: $e_p = p_d - p,\quad e_v = v_d - v$

偏航前馈限幅 → ⑩: $\dot\psi_c = \text{clip}(\dot\psi_d, \pm\dot\psi_{\max})$，$\ddot\psi_c = \begin{cases} 0, & |\dot\psi_d| > \dot\psi_{\max} \text{ 或 } (|\dot\psi_d| = \dot\psi_{\max} \wedge \dot\psi_d\ddot\psi_d > 0) \\ \ddot\psi_d, & \text{否则} \end{cases}$

积分（上限 $\bar\eta$；$k_{i,j} = 0$ 的轴恒为 0）: $\tilde\eta = \text{clip}(\eta + e_p\,\Delta t_c,\ \pm\bar\eta),\quad \tilde\eta_j = 0$ 若 $k_{i,j} = 0$

抗饱和：上一步分配受限（⑭ 反馈）时只许变小: $\ell_{\text{alloc}} \Rightarrow \eta_j^+ = \begin{cases} \tilde\eta_j, & |\tilde\eta_j| \le |\eta_j| \\ \eta_j, & \text{否则} \end{cases}$

期望加速度: $a_c = \underbrace{K_p e_p}_{\text{比例}} + \underbrace{K_v e_v}_{\text{阻尼}} + \underbrace{a_d}_{\text{前馈}} + \underbrace{K_i \eta^+}_{\text{积分}}$

限幅 1：加速度范数；再换成推力向量: $a' = a_c \min\Big(1, \frac{a_{\max}}{\|a_c\|}\Big),\quad F = m\,(a' - g)$

限幅 2：正交投影到半角 $\beta_{\max}$ 的推力圆锥（含锥顶）$\Pi(\cdot)$: $F \leftarrow \begin{cases} F, & F_z \ge 0 \wedge \|F_{xy}\| \cos\beta_{\max} \le F_z \sin\beta_{\max} \\ \max(0,\ \sin\beta_{\max}\|F_{xy}\| + \cos\beta_{\max} F_z) \begin{bmatrix} \sin\beta_{\max} F_{xy}/\|F_{xy}\| \\ \cos\beta_{\max} \end{bmatrix}, & \text{否则} \end{cases}$

限幅标志：回滚增长的积分并重算: $\ell = [\|a_c\| > a_{\max}] \vee [\text{锥外}]$，$\ell \Rightarrow \eta_j \leftarrow \eta_j^{\text{prev}}\ (|\eta_j| \uparrow),\quad F = \Pi(m(a' - g))$

推力中的反馈部分 → ⑨: $F_{\text{fb}} = F - m a_d$

$F,\ F_{\text{fb}},\ \ell$ → ⑨

F → ⑪

9 期望推力的导数

`geometric.py: _force_derivatives`

变步长后向差分（最近三次历史 x_k, x_{k-1}, x_{k-2}）: $\delta_k = \frac{x_k - x_{k-1}}{\Delta t_k},\quad \delta_{k-1} = \frac{x_{k-1} - x_{k-2}}{\Delta t_{k-1}}$

中点二阶差分，外推半步到当前时刻: $\mathcal{D}_2[x] = \frac{2(\delta_k - \delta_{k-1})}{\Delta t_k + \Delta t_{k-1}},\quad \mathcal{D}_1[x] = \delta_k + \frac{1}{2}\Delta t_k\,\mathcal{D}_2[x]$

历史不足（复位后）: 无历史：$\mathcal{D}_1 = \mathcal{D}_2 = 0$；一个：$\mathcal{D}_1 = \delta_k,\ \mathcal{D}_2 = 0$

未限幅：参考解析 + 反馈数值: $\neg\ell_k \wedge \neg\ell_{k-1} \wedge \neg\ell_{k-2}:$ $\dot F = m j_d + \mathcal{D}_1[F_{\text{fb}}]$，$\ddot F = m s_d + \mathcal{D}_2[F_{\text{fb}}]$

限幅：总推力整体差分: 否则：$\dot F = \mathcal{D}_1[F]$，$\ddot F = \mathcal{D}_2[F]$

$\dot F,\ \ddot F$ → ⑩

10 期望姿态运动学 $\mathcal{K}$

`rotations.py: desired_attitude_kinematics`

单位方向及其导数 $\mathcal{N}(x, \dot x, \ddot x)$（链式法则）: $u = \frac{x}{\|x\|},\quad \dot\ell = u^\top\dot x,\quad \dot u = \frac{\dot x - u\dot\ell}{\|x\|},\quad \ddot\ell = \dot u^\top\dot x + u^\top\ddot x,\quad \ddot u = \frac{\ddot x - u\ddot\ell - 2\dot u\dot\ell}{\|x\|}$

机頂对准推力: $(b_3, \dot b_3, \ddot b_3) = \mathcal{N}(F, \dot F, \ddot F)$

水平航向（$\psi_c, \dot\psi_c$ 来自 ⑧）: $h = (\cos\psi_d, \sin\psi_d, 0)^\top,\quad h_\perp = (-\sin\psi_d, \cos\psi_d, 0)^\top$，$\dot h = h_\perp\dot\psi_c,\quad \ddot h = h_\perp\ddot\psi_c - h\,\dot\psi_c^2$

左翼方向（未归一化）叉乘的乘法求导: $c = b_3 \times h,\quad \dot c = \dot b_3 \times h + b_3 \times \dot h,\quad \ddot c = \ddot b_3 \times h + 2\dot b_3 \times \dot h + b_3 \times \ddot h$

归一化: $(b_2, \dot b_2, \ddot b_2) = \mathcal{N}(c, \dot c, \ddot c)$

机头方向: $b_1 = b_2 \times b_3,\quad \dot b_1 = \dot b_2 \times b_3 + b_2 \times \dot b_3,\quad \ddot b_1 = \ddot b_2 \times b_3 + 2\dot b_2 \times \dot b_3 + b_2 \times \ddot b_3$

期望姿态及其导数: $R_d = [b_1\ b_2\ b_3],\quad \dot R_d = [\dot b_1\ \dot b_2\ \dot b_3],\quad \ddot R_d = [\ddot b_1\ \ddot b_2\ \ddot b_3]$

由 $\dot R = R\hat\Omega$（⓪）反推；$\dot R^\top\dot R$ 对称，被 vee 消去: $\Omega_d = (R_d^\top\dot R_d)^\vee,\quad \dot\Omega_d = (\dot R_d^\top\dot R_d + R_d^\top\ddot R_d)^\vee$

退化分支: $\|F\| < \epsilon:\quad R_d = R_d^{\text{prev}},\quad \Omega_d = \dot\Omega_d = 0$；$\|b_3 \times h\| < \epsilon:\quad h \leftarrow R_d^{\text{prev}} e_1$（或与 b_3 最不平行的坐标轴），$\dot h = \ddot h = 0$；$R_d^{\text{prev}} = R$（首次调用）

$R_d,\ \Omega_d,\ \dot\Omega_d$ → ⑪

ℓ_{alloc}（抗饱和，下一控制步）→ ⑧

11 Lee SO(3) 姿态控制律

`control/geometric.py`

期望姿态在当前机体系中的表示: $R_e = R^\top R_d$

姿态误差（不经欧拉角）: $e_R = \frac{1}{2}(R_e^\top - R_e)^\vee = \frac{1}{2}(R_d^\top R - R^\top R_d)^\vee$

角速度误差: $e_\Omega = \Omega - R_e\Omega_d$

角加速度前馈: $\alpha_{\text{ff}} = R_e\dot\Omega_d - \Omega \times (R_e\Omega_d)$

力矩：增益直接出力矩（不再乘惯量）: $M_d = \underbrace{-K_R e_R}_{\text{比例}}\ \underbrace{-K_\Omega e_\Omega}_{\text{阻尼}} + \underbrace{\Omega \times J\Omega}_{\text{陀螺补偿}} + \underbrace{J\alpha_{\text{ff}}}_{\text{角加速度前馈}}$

机体系力旋量（零阶保持 N_c 个物理步）: $w_d = (0,\ 0,\ f_\Sigma,\ M_d) \in \mathbb{R}^6,\quad w_d(t) = w_d(t_{k_c}),\quad t \in [t_{k_c}, t_{k_c} + \Delta t_c)$

总推力 = F 在当前机顶上的投影: $f_\Sigma = \max(0,\ F^\top R e_3)$

w_d → ⑬

12 级联 PID（对照基线，可整体替换模块 8–11）

`control/cascaded_pid.py`

位置环 → 速度设定（修正量范数限幅）: $\Delta v = \text{clip}_{\|\cdot\| \le \bar e_v}(K_{pp} e_p),\quad v_{sp} = v_d + \Delta v,\quad e_v' = v_{sp} - v$

速度环 → 加速度: $a = a_d + K_{pv} e_v' + I_v,\quad (F, \ell) = \Pi(a)$（同 ⑧ 限幅）

速度积分（ℓ 时只减不增）: $I_v^+ = \text{clip}(I_v + K_{iv} e_v' \Delta t_c,\ \pm\bar I_v)$

期望姿态（不用 jerk/snap）: $(R_d, \Omega_d) = \mathcal{K}(F,\ 0,\ 0,\ \psi_d,\ \dot\psi_c,\ 0)$

姿态环 → 角速度设定: $\Omega_{sp} = \text{clip}(-K_a e_R + R_e\Omega_d,\ \pm\bar\Omega)$

分配残差 ⇒ 两个积分都回滚增长分量: $\|\varrho\| > \epsilon \Rightarrow I_j \leftarrow I_j^{\text{prev}}\ (|I_j| \uparrow)$

角速度环（按惯量归一化，增益即带宽）: $e_\omega = \Omega_{sp} - \Omega,\quad \alpha = K_{pr} e_\omega + I_r,\quad I_r^+ = \text{clip}(I_r + K_{ir} e_\omega \Delta t_c,\ \pm\bar I_r)$

输出（与 ⑪ 相同接口）: $M = J\alpha + \Omega \times J\Omega,\quad f_\Sigma = \max(0,\ F^\top R e_3),\quad w_d = (0, 0, f_\Sigma, M)$

w_d（可替换）→ ⑬

13 有界加权最小二乘推力分配 $\mathcal{A}$

`control/allocation.py`

第 i 个旋翼对整机质心的贡献（实际几何）: $F_i = f_i a_i,\quad M_i = (r_i - c) \times f_i a_i + \sigma_i k_{\tau,i} f_i a_i,\quad a_i = e_3,\ \sigma_i \in \{\pm1\}$

分配矩阵: $A_i = \begin{bmatrix} a_i \\ (r_i - c) \times a_i + \sigma_i k_{\tau,i} a_i \end{bmatrix},\quad w = A\mathbf{f} = \sum_i A_i f_i,\quad A \in \mathbb{R}^{6\times4}$

上下限（电机 ∩ 电池 ⑱）: $\underline{\mathbf{f}} = \max(\mathbf{f}_{\min}, \underline{\mathbf{f}}_P)$，$\bar{\mathbf{f}} = \min(\mathbf{f}_{\max}, \bar{\mathbf{f}}_P)$

优化问题（$W = \text{diag}(\text{weights})$，$\lambda \ge 0$ 平滑惩罚，堆叠成单个最小二乘）: $\mathbf{f}^\star = \arg\min_{\underline{\mathbf{f}} \le \mathbf{f} \le \bar{\mathbf{f}}} \|W(A\mathbf{f} - w_d)\|^2 + \lambda\|\mathbf{f} - \mathbf{f}_{\text{prev}}\|^2 = \arg\min_{\underline{\mathbf{f}} \le \mathbf{f} \le \bar{\mathbf{f}}} \|\tilde A\mathbf{f} - \tilde b\|^2$，$\tilde A = \begin{bmatrix} WA \\ \sqrt\lambda I \end{bmatrix},\quad \tilde b = \begin{bmatrix} Ww_d \\ \sqrt\lambda\,\mathbf{f}_{\text{prev}} \end{bmatrix}$

枚举活动集（$\underline{f}_i = \bar f_i$ 时只有一种）: $s \in \{-1, 0, +1\}^4\ (\le 3^4$ 种$),\quad f_i = \begin{cases} \underline{f}_i, & s_i = -1 \\ \bar f_i, & s_i = +1 \end{cases},\quad \mathbf{f}_{\mathcal F} = \tilde A_{\mathcal F}^+(\tilde b - \tilde A_{\bar{\mathcal F}} \mathbf{f}_{\bar{\mathcal F}}),\quad \mathcal F = \{i : s_i = 0\}$

取可行且代价最小者: $\underline{\mathbf{f}} - \epsilon \le \mathbf{f}(s) \le \bar{\mathbf{f}} + \epsilon,\quad J(s) = \|\tilde A\mathbf{f}(s) - \tilde b\|^2,\quad \mathbf{f}^\star = \mathbf{f}(\arg\min_s J(s))$

残差与饱和标志（写入日志）: $w_{\text{ach}} = A\mathbf{f}^\star,\quad \varrho = w_d - w_{\text{ach}},\quad \text{sat} = \exists i:\ f_i^\star \in \{\underline{f}_i, \bar f_i\}$

直接输出推力的控制器: $\mathbf{f}^\star = \text{clip}(\mathbf{f}_c,\ \underline{\mathbf{f}},\ \bar{\mathbf{f}})$

$\mathbf{f}^\star$ → ⑭

14 推力 → 转速、起转包络

`actuation.py / runtime/loop.py`

推力请求 → 转速（rps；零指令保持为零）: $\hat f_i = \begin{cases} 0, & f_i^\star = 0 \\ \text{clip}(f_i^\star,\ f_{\min},\ f_{\max}), & f_i^\star > 0 \end{cases},\quad n_i = \sqrt{\hat f_i / k_{f,i}}$

单位换算（转子转速 ≠ 机体角速度）: $\text{rpm} = 60\,n,\quad \omega_r = 2\pi n,\quad k_\omega = \frac{k_f}{(2\pi)^2}$

起转包络（同一个九次 S，①）: $\tilde n_i = \begin{cases} S(t/T_s)\,n_i, & t < T_s\ (\text{spin\_up}) \\ n_i, & \text{否则} \end{cases}$

运行上下限: $n_{\text{tgt},i} = \begin{cases} 0, & \tilde n_i = 0 \\ \text{clip}(\tilde n_i,\ n_{i,\min},\ n_{i,\max}), & \tilde n_i > 0 \end{cases},\quad n_{i,\min/\max} = \sqrt{f_{\min/\max}/k_{f,i}}$

实际可请求的推力与力旋量: $f_{i,\text{eff}} = k_{f,i}\,n_{\text{tgt},i}^2,\quad w_{\text{eff}} = A\,\mathbf{f}_{\text{eff}},\quad \mathbf{f}_{\text{prev}} \leftarrow \mathbf{f}_{\text{eff}}$

抗饱和反馈 → ⑧ ⑫: $\varrho_{\text{eff}} = w_d - w_{\text{eff}},\quad \ell_{\text{alloc}} = [\|\varrho_{\text{eff}}\| > \epsilon]$

n_{tgt} → ⑮

15 非对称一阶电机动态 $\mathcal{M}$

`actuation.py / native_thruster.py`

升速 / 降速时间常数不同: $\tau_i = \begin{cases} \tau_{\text{dec},i}, & n_{\text{tgt},i} < n_i \\ \tau_{\text{inc},i}, & \text{否则} \end{cases}$

混合系数: $\alpha_i = \begin{cases} 1/(\Delta t + \tau_i), & \text{离散近似} \\ 1/\tau_i, & \text{连续} \end{cases}$

转速变化率（q 限制 rps/s）: $G_i(e) = \text{clip}(\alpha_i e,\ -q,\ q),\quad e_i = n_{\text{tgt},i} - n_i,\quad \dot n_i = G_i(e_i)$

Euler: $n_i^+ = n_i + \Delta t\,G_i(e_i)$

RK4（本步 τ_i, α_i 与目标固定）: $k_1 = G(e),\quad k_2 = G(e - \frac{1}{2}\Delta t\,k_1),\quad k_3 = G(e - \frac{1}{2}\Delta t\,k_2),\quad k_4 = G(e - \Delta t\,k_3)$，$n^+ = n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

状态限幅；由实际转速产生实际推力: $n_i^+ \leftarrow \text{clip}(n_i^+,\ 0,\ n_{i,\max}),\quad f_{i,\text{act}} = k_{f,i}(n_i^+)^2$

电机力旋量（诊断，不重复施加）: $w_{\text{motor}} = A\,\mathbf{f}_{\text{act}}$

$\mathbf{f}_{\text{act}}$ → ⑰

$[\underline{\mathbf{f}}_P, \bar{\mathbf{f}}_P]$ → ⑬

16 风场、气动阻力与外部扰动

`effects/*.py`

OU 湍流（精确离散化）: $\xi^+ = e^{-\Delta t/\tau_w}\xi + \sigma_w\sqrt{1 - e^{-2\Delta t/\tau_w}}\,\mathcal{N}(0, I)$

半余弦阵风脉冲: $g_k(t) = \begin{cases} \frac{1}{2}(1 - \cos 2\pi\phi_k)\,\Delta v_k, & 0 < \phi_k < 1 \\ 0, & \text{否则} \end{cases},\quad \phi_k = \frac{t - t_k}{T_k}$

世界系风速场（仿射梯度）: $u_w(x, t) = \bar u + \xi + G_w(x - x_{\text{ref}}) + \sum_k g_k(t)$

压心点的位置与速度（机体系）: $x_{cp} = p + R\,r_{cp},\quad v_{cp}^B = R^\top v + \Omega \times r_{cp}$

相对气流: $v_{\text{rel}} = v_{cp}^B - R^\top u_w(x_{cp}, t)$

集总阻力（D, D_Ω 对称部分半正定）: $F_{\text{drag}} = \underbrace{-D\,v_{\text{rel}}}_{\text{线性}}\ \underbrace{-q \odot |v_{\text{rel}}| \odot v_{\text{rel}}}_{\text{二次}},\quad M_{\text{drag}} = r_{cp} \times F_{\text{drag}} - D_\Omega\Omega$

恒定力 / 力偶，$t \in [t_s, t_e)$: $F^B = \begin{cases} F & \text{body} \\ R^\top F & \text{world} \end{cases},\quad M^B = \tau^B + r^B \times F^B,\quad r^B = \begin{cases} r_b \\ R^\top(x_w - p) \end{cases}$

外部力旋量（机体系，关于质心）: $w_{\text{ext}} = \sum_{\text{效应}}(F^B, M^B) = (F_{\text{drag}} + \sum F^B,\ M_{\text{drag}} + \sum M^B)$

w_{ext} → ⑰

$x_{k+1} \to x_k$（下一步）

17 刚体动力学（Newton–Euler + RK4）$\Phi_{\Delta t}$

`sim/synthetic.py`

提交到刚体的力旋量（机体系，关于质心）: $w_{\text{sub}} = A\,\mathbf{f}_{\text{act}} + w_{\text{ext}} = (F^B,\ M^B)$

平动（世界系）: $\dot p = v,\quad \dot v = g + \frac{1}{m} R F^B$

四元数运动学: $\dot q = \frac{1}{2} q \otimes \begin{bmatrix} 0 \\ \Omega \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -q_v^\top\Omega \\ q_w\Omega + q_v \times \Omega \end{bmatrix}$

转动（欧拉方程，完整惯量）: $\dot\Omega = J^{-1}(M^B - \Omega \times J\Omega)$

13 维状态的经典 RK4: $y = (p, v, q, \Omega),\quad k_1 = f(y),\quad k_2 = f(y + \frac{\Delta t}{2}k_1),\quad k_3 = f(y + \frac{\Delta t}{2}k_2),\quad k_4 = f(y + \Delta t\,k_3)$，$y^+ = y + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4),\quad q^+ \leftarrow q^+/\|q^+\|$

单边无摩擦质心支撑（非弹性冲量）: $z < z_{\text{floor}} \Rightarrow z \leftarrow z_{\text{floor}};\quad v_z < 0 \Rightarrow J_c = -m\,v_z,\quad v_z \leftarrow 0$

下一时刻状态: $x_{k+1} = (p, v, q, \Omega)^+,\quad t_{k+1} = t_k + \Delta t$

根链接 ↔ 整机质心（c 为机体系质心）: $p = p_{\text{root}} + Rc,\quad v = v_{\text{root}} + \omega_W \times Rc,\quad \Omega = R^\top\omega_W$

Isaac Lab 后端：PhysX 积分: 各桨 $f_{i,\text{act}} a_i$ 与反扭矩 $\sigma_i k_{\tau,i} f_{i,\text{act}} a_i$ 施加在各旋翼原点，经固定关节传给整机；外力移到根刚体质心：$M_{\text{root}} = M_{\text{ext}} + (c - c_{\text{root}}) \times F_{\text{ext}}$

控制分频: $k \bmod N_c = 0 \Rightarrow$ 更新 $\mathcal{S}(t_k),\ w_d;\quad \Delta t_c = N_c\,\Delta t$

$x_k,\ x_{k+1}$ → ⑲

x_{k+1} → ⑳

18 电池等效电路（可选）

`power/battery.py`

库仑效率与 SOC（电流在步内恒定）: $I_{\text{eff}} = \begin{cases} I/\eta_d, & I \ge 0\ (\text{放电}) \\ I\,\eta_c, & I < 0 \end{cases},\quad \text{SOC}^+ = \text{SOC} - \frac{I_{\text{eff}}\,\Delta t}{3600\,Q_{\text{Ah}}}$

RC 极化支路（对恒流精确）: $\tau_j = R_j C_j,\quad x_j = \frac{\Delta t}{\tau_j},\quad \Delta V_j = IR_j - V_j$，$V_j^+ = V_j + \Delta V_j(1 - e^{-x_j})$

开路电压（分段线性插值）: $V_{\text{oc}}(\text{SOC}) = \text{interp}(\text{SOC};\ \text{SOC}_k, V_k)$

端电压: $V_t = \underbrace{V_{\text{oc}}(\text{SOC}^+)}_{\text{开路}}\ \underbrace{-IR_0}_{\text{欧姆}}\ \underbrace{-\sum_j V_j^+}_{\text{极化}}$

标定的推力包络 → ⑬: $[\underline{\mathbf{f}}_P, \bar{\mathbf{f}}_P] = \mathcal{E}(\text{电池状态},\ x,\ \text{遥测})$，$V_t \le V_{\text{cut}} \vee V_t \ge V_{\max} \vee \text{SOC} \le 0 \Rightarrow$ 拒绝授权

输出能量（端电压精确积分）: $E^+ = E + I\int_0^{\Delta t} V_t\,dt,\quad \int V_t\,dt = \Delta t\,\overline{V_{\text{oc}}} - IR_0\Delta t - \sum_j[V_j\Delta t + \Delta V_j\,a_j]$，$a_j = \Delta t - \tau_j(1 - e^{-x_j}),\quad \overline{V_{\text{oc}}} = \text{SOC}$ 区间上 V_{oc} 的梯形均值

焦耳热与电荷吞吐: $Q^+ = Q + I^2 R_0\Delta t + \sum_j \frac{V_j^2\Delta t + 2V_j\Delta V_j a_j + \Delta V_j^2 b_j}{R_j}$，$b_j = \Delta t - 2\tau_j(1 - e^{-x_j}) + \frac{\tau_j}{2}(1 - e^{-2x_j}),\quad C^+ = C + |I|\Delta t$

19 每个物理步的遥测派生量

`telemetry/measurements.py`

输入：区间两端的实际状态与本区间保持的控制目标: $x_k = (p, v, R, \Omega),\quad x_{k+1} = (p^+, v^+, R^+, \Omega^+),\quad \mathcal{S},\ R_d,\ \Omega_d,\ w_d,\ n_{\text{tgt}}$

区间平均加速度与净力: $\Delta t = t^+ - t,\quad \hat a = \frac{v^+ - v}{\Delta t},\quad F_{\text{net}} = m\,\hat a$

角加速度（先转世界系再差分）: $\omega_W = R\Omega,\quad \hat\alpha_W = \frac{R^+\Omega^+ - R\Omega}{\Delta t},\quad \hat\alpha_B = R^{+\top}\hat\alpha_W$

净力矩 = 世界系角动量差分: $L_W = RJ\Omega,\quad M_{\text{net}} = \frac{L_W^+ - L_W}{\Delta t}$

跟踪误差: $e_p = p_d - p^+,\quad e_v = v_d - v^+,\quad e_a = a_d - \hat a$

姿态误差角（⓪ 对数映射）: $\vartheta_e = \|\operatorname{Log}(R^{+\top} R_d)\|,\quad e_{\text{rpy}} = \text{wrap}(\text{rpy}(R_d) - \text{rpy}(R^+))$

角速度误差（换到终点机体系）: $e_\Omega = R^{+\top} R_d\Omega_d - \Omega^+$

电机与力旋量误差: $e_n = n_{\text{tgt}} - n^+,\quad e_f = k_f n_{\text{tgt}}^2 - \mathbf{f}_{\text{act}},\quad e_w = w_d - w_{\text{motor}}$

20 完成判据与运行指标

`completion.py / analysis/metrics.py`

偏航误差（最短角差）: $\psi = \text{atan2}(R_{21}, R_{11}),\quad e_\psi = \text{atan2}(\sin(\psi_d - \psi),\ \cos(\psi_d - \psi))$

容差指示: $\chi(t) = [\text{hold}] \wedge [\|p_d - p\| \le \varepsilon_p] \wedge [\|v\| \le \varepsilon_v] \wedge [|e_\psi| \le \varepsilon_\psi] \wedge [\|\Omega\| \le \varepsilon_\Omega]$

连续驻留: $t_\chi = \inf\{s:\ \chi \equiv 1$ 于 $[s, t]\},\quad \text{dwell} = t - t_\chi,\quad$ 达成 $= \chi \wedge (\text{dwell} \ge T_{\text{dwell}})$

成功 / 超时: 成功 $\iff t_\chi + T_{\text{dwell}} \le t_{\text{hold}} + T_{\max},\quad T_{\text{sim}} \ge b_{\text{end}} + T_{\text{dwell}} + 2\Delta t$

误差统计（全程 / 各阶段）: $\text{RMS} = \sqrt{\frac{1}{N}\sum_k \|e_k\|^2},\quad \max_k \|e_k\|,\quad \|e_N\|$

姿态与执行器指标: $\beta_k = \arccos(1 - 2(q_x^2 + q_y^2)) = \arccos R_{33}$，饱和比例 $= \frac{1}{N}\sum_k \mathbf{1}[\text{sat}_k],\quad \bar f_\Sigma = \frac{1}{N}\sum_k\sum_i f_{i,k}$